

Development of the Ideal Op Amp Equations

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3.1 Ideal Op Amp Assumptions

The name *ideal op amp* is applied to this and similar analyses because the salient parameters of the op amp are assumed to be perfect. An engineer may wish that an ideal op amp existed at times, but if such a component actually did exist, it would destroy the known universe! See the end of this chapter for an explanation. We are all thankful that there is no such thing as an ideal op amp, but present day op amps come so close to ideal that *ideal op amp* analysis approaches actual analysis. Op amps depart from the ideal in two ways. First, DC parameters, such as input offset voltage, are large enough to cause departure from the ideal. The ideal assumes that input offset voltage is zero. Second, AC parameters, such as gain, are a function of frequency, so they go from large values at DC to small values at high frequencies.

This assumption simplifies the analysis, thus it clears the path for insight. It is so much easier to see the forest when the brush and huge trees are cleared away. Although the ideal op amp analysis makes use of perfect parameters, the analysis is often valid because some op amps approach perfection. In addition, when working at low frequencies, several kilohertz, the ideal op amp analysis produces accurate answers. Voltage feedback op amps are covered in this chapter, and current feedback op amps are covered in Chapter 9.

Several assumptions have to be made before the ideal op amp analysis can proceed. First, assume that the current flow into the input leads of the op amp is zero. This

assumption is almost true in FET op amps where input currents can be less than a picoamp, but this is not always true in bipolar high speed op amps, where tens of microamp input currents are found.

Second, the op amp gain is assumed to be infinite, hence it drives the output voltage to any value to satisfy the input conditions. This assumes that the op amp output voltage can achieve any value. In reality, saturation occurs when the output voltage comes close to a power supply rail; but reality does not negate the assumption, it only bounds it.

Also, implicit in the infinite gain assumption is the need for zero input signal. The gain drives the output voltage until the voltage between the input leads (the error voltage) is zero. This leads to the third assumption, that the voltage between the input leads is zero. The implication of zero voltage between the input leads means that, if one input is tied to a hard voltage source such as ground, then the other input is at the same potential. The current flow into the input leads is zero, so the input impedance of the op amp is infinite.

Fourth, the output impedance of the ideal op amp is zero. The ideal op amp can drive any load without an output impedance dropping voltage across it. The output impedance of most op amps is a fraction of an ohm for low current flows, so this assumption is valid in most cases. Fifth, the frequency response of the ideal op amp is flat: This means that the gain does not vary as frequency increases. By constraining the use of the op amp to the low frequencies, we make the frequency response assumption true.

[Table 3.1](#) lists the basic ideal op amp assumptions and [Figure 3.1](#) shows the ideal op amp.

Table 3.1: Basic Ideal Op Amp Assumptions

Parameter name	Parameters symbol	Value
Input current	I_{IN}	0
Input offset voltage	V_{OS}	0
Input impedance	Z_{IN}	∞
Output impedance	Z_{OUT}	0
Gain	a	∞

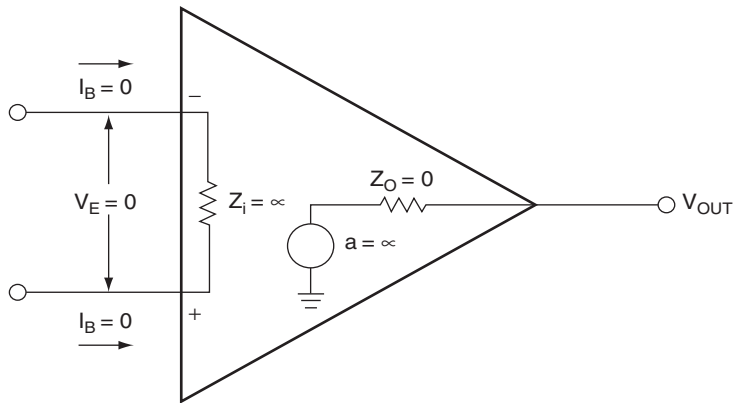


Figure 3.1: The ideal op amp.

3.2 The Noninverting Op Amp

The noninverting op amp has the input signal connected to its noninverting input (Figure 3.2), thus its input source sees an infinite impedance. There is no input offset voltage, because $V_{OS} = V_E = 0$, hence the negative input must be at the same voltage as the positive input. The op amp output drives current into R_F until the negative input is at the voltage, V_{IN} . This action causes V_{IN} to appear across R_G .

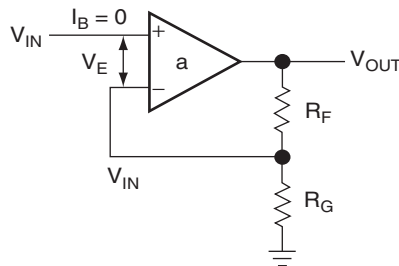


Figure 3.2: The noninverting op amp.

The voltage divider rule is used to calculate V_{IN} : V_{OUT} is the input to the voltage divider, and V_{IN} is the output of the voltage divider. Since no current can flow into either op amp lead, use of the voltage divider rule is allowed. Equation (3.1)

is written with the aid of the voltage divider rule, and algebraic manipulation yields Equation (3.2) in the form of a gain parameter:

$$V_{\text{IN}} = V_{\text{OUT}} \frac{R_G}{R_G + R_F} \quad (3.1)$$

$$\frac{V_{\text{OUT}}}{V_{\text{IN}}} = \frac{R_G + R_F}{R_G} = 1 + \frac{R_F}{R_G} \quad (3.2)$$

when R_G becomes very large with respect to R_F , $(R_F/R_G) \Rightarrow 0$ and Equation (3.2) reduces to Equation (3.3):

$$V_{\text{OUT}} = 1 \quad (3.3)$$

Under these conditions $V_{\text{OUT}} = 1$ and the circuit becomes a unity gain buffer. R_G is usually deleted to achieve the same results, and when R_G is deleted, R_F can also be deleted (R_F must be shorted when it is deleted). When R_F and R_G are deleted, the op amp output is connected to its inverting input with a wire. Some op amps are self-destructive when R_F is left out of the circuit, so R_F is used in many buffer designs. When R_F is included in a buffer circuit, its function is to protect the inverting input from an overvoltage to limit the current through the input ESD (electrostatic discharge) structure (typically <1 mA), and it can have almost any value (20 k is often used). R_F can never be left out of the circuit in a current feedback amplifier design because R_F determines stability in current feedback amplifiers.

3.3 The Inverting Op Amp

The noninverting input of the inverting op amp circuit is grounded (Figure 3.3). One assumption made is that the input error voltage is zero, so the feedback keeps inverting the input of the op amp at a virtual ground (not actual ground but acting like ground). The current flow in the input leads is assumed to be zero, hence the current flowing through R_G equals the current flowing through R_F . Using Kirchhoff's law, we write Equation (3.4); and the minus sign is inserted because this is the inverting input. Algebraic manipulation gives Equation (3.5).

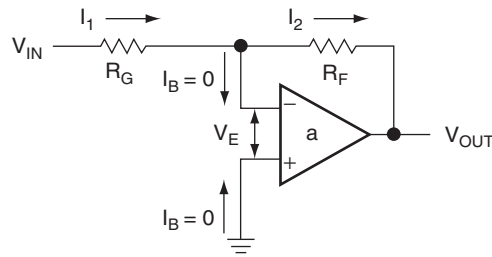


Figure 3.3: The inverting op amp.

$$I_1 = \frac{V_{IN}}{R_G} = -I_2 = -\frac{V_{OUT}}{R_F} \quad (3.4)$$

$$\frac{V_{OUT}}{V_{IN}} = -\frac{R_F}{R_G} \quad (3.5)$$

Note that the gain is a function of only the feedback and gain resistors, so the feedback has accomplished its function of making the gain independent of the op amp parameters. The actual resistor values are determined by the impedance levels that the designer wants to establish. If $R_F = 10 \text{ k}$ and $R_G = 10 \text{ k}$, the gain is -1 , as shown in Equation (3.5); and if $R_F = 100 \text{ k}$ and $R_G = 100 \text{ k}$, the gain is still -1 . The impedance levels of 10 k or 100 k determine the current drain, the effect of stray capacitance, and a few other points. The impedance level does not set the gain; the ratio of R_F/R_G does.

One final note: The output signal is the input signal amplified and inverted. The circuit input impedance is set by R_G , because the inverting input is held at a virtual ground.

3.4 The Adder

An adder circuit can be made by connecting more inputs to the inverting op amp (Figure 3.4). The opposite end of the resistor connected to the inverting input is held at virtual ground by the feedback; therefore, adding new inputs does not affect the response of the existing inputs.

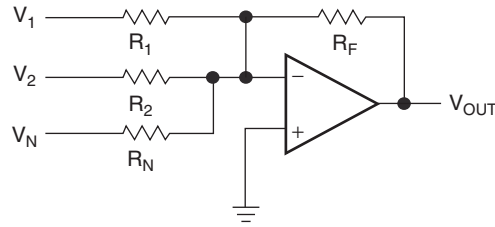


Figure 3.4: The adder circuit.

Superposition is used to calculate the output voltages resulting from each input, and the output voltages are added algebraically to obtain the total output voltage. Equation (3.6) is the output equation when V_1 and V_2 are grounded. Equations (3.7) and (3.8) are the other superposition equations, and the final result is given in Equation (3.9).

$$V_{OUTN} = -\frac{R_F}{R_N} V_N \quad (3.6)$$

$$V_{OUT1} = -\frac{R_F}{R_1} V_1 \quad (3.7)$$

$$V_{OUT2} = -\frac{R_F}{R_2} V_2 \quad (3.8)$$

$$V_{OUT} = -\left(\frac{R_F}{R_1} V_1 + \frac{R_F}{R_2} V_2 + \frac{R_F}{R_N} V_N\right) \quad (3.9)$$

3.5 The Differential Amplifier

The differential amplifier circuit amplifies the difference between signals applied to the inputs (Figure 3.5). Superposition is used to calculate the output voltage resulting from each input voltage, then the two output voltages are added to arrive at the final output voltage.

The op amp input voltage resulting from the input source, V_1 , is calculated in Equations (3.10) and (3.11). The voltage divider rule is used to calculate the voltage,

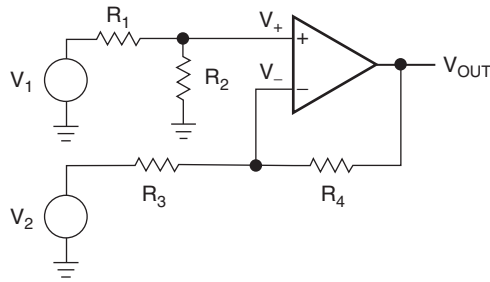


Figure 3.5: The differential amplifier.

V_+ , and the noninverting gain equation, [Equation \(3.2\)](#), is used to calculate the noninverting output voltage, V_{OUT1} :

$$V_+ = V_1 \frac{R_2}{R_1 + R_2} \quad (3.10)$$

$$V_{OUT1} = V_+(G_+) = V_1 \frac{R_2}{R_1 + R_2} \left(\frac{R_3 + R_4}{R_3} \right) \quad (3.11)$$

The inverting gain equation, [Equation \(3.5\)](#), is used to calculate the stage gain for V_{OUT2} in [Equation \(3.12\)](#). These inverting and noninverting gains are added in [Equation \(3.13\)](#).

$$V_{OUT2} = V_2 \left(\frac{-R_4}{R_3} \right) \quad (3.12)$$

$$V_{OUT} = V_1 \frac{R_2}{R_1 + R_2} \left(\frac{R_3 + R_4}{R_3} \right) - V_2 \frac{R_4}{R_3} \quad (3.13)$$

$$V_{OUT} = (V_1 - V_2) \frac{R_4}{R_3} \quad (3.14)$$

It is now obvious that the differential signal, $(V_1 - V_2)$, is multiplied by the stage gain, so the name *differential amplifier* suits the circuit. Because it amplifies only the differential portion of the input signal, it rejects the common mode portion of

the input signal. A common mode signal is illustrated in [Figure 3.6](#). Because the differential amplifier strips off or rejects the common mode signal, this circuit configuration is often employed to strip DC or injected common mode noise off a signal.

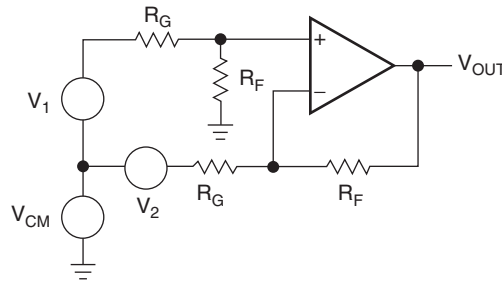


Figure 3.6: Differential amplifier with common mode input signal.

The disadvantage of this circuit is that the two input impedances cannot be matched when it functions as a differential amplifier, thus the two or three op amp versions of this circuit specially designed for high performance applications require matched input impedances.

3.6 Complex Feedback Networks

When complex networks are put into the feedback loop, the circuits get harder to analyze, because the simple gain equations cannot be used. The usual technique is to write and solve node or loop equations. There is only one input voltage, so superposition is not of use, but Thevenin's theorem can be used, as is shown in the example problem that follows.

Sometimes, it is desirable to have a low resistance path to ground in the feedback loop. Standard inverting op amps cannot do this when the driving circuit sets the input resistor value and the gain specification sets the feedback resistor value. Inserting a T network in the feedback loop ([Figure 3.7](#)) yields a degree of freedom that enables both specifications to be met with a low DC resistance path in the feedback loop.

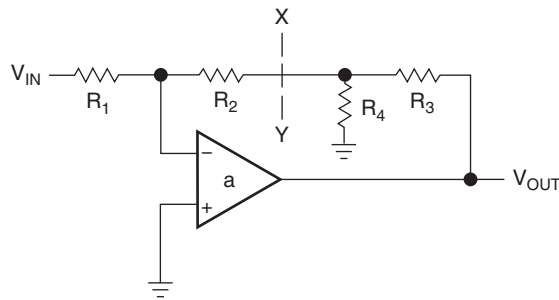


Figure 3.7: T network in feedback loop.

Break the circuit at point X – Y , stand on the terminals looking into R_4 , and calculate the Thevenin equivalent voltage as shown in Equation (3.15). The Thevenin equivalent impedance is calculated in Equation (3.16).

$$V_{TH} = V_{OUT} \frac{R_4}{R_3 + R_4} \quad (3.15)$$

$$R_{TH} = R_3 \parallel R_4 \quad (3.16)$$

Replace the output circuit with the Thevenin equivalent circuit, as shown in Figure 3.8, and calculate the gain with the aid of the inverting gain equation as shown in Equation (3.17):

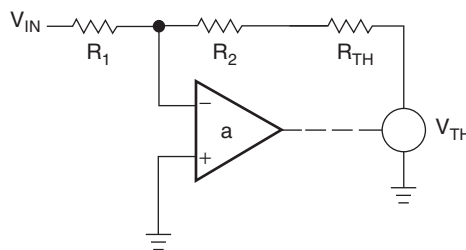


Figure 3.8: Thevenin's theorem applied to a T network.

$$-\frac{V_{TH}}{V_{IN}} = \frac{R_2 + R_{TH}}{R_1} \quad (3.17)$$

Substituting the Thevenin equivalents into Equation (3.17) yields Equation (3.18):

$$-\frac{V_{\text{OUT}}}{V_{\text{IN}}} = \frac{R_2 + R_{\text{TH}}}{R_1} \left(\frac{R_3 + R_4}{R_4} \right) = \frac{R_2 + (R_3 \parallel R_4)}{R_1} \left(\frac{R_3 + R_4}{R_4} \right) \quad (3.18)$$

Algebraic manipulation yields Equation (3.19):

$$-\frac{V_{\text{OUT}}}{V_{\text{IN}}} = \frac{R_2 + R_3 + \frac{R_2 R_3}{R_4}}{R_1} \quad (3.19)$$

Specifications for the circuit you are required to build are an inverting amplifier with an input resistance of 10 k ($R_G = 10$ k), a gain of 100, and a feedback resistance of 20 k or less. The inverting op amp circuit cannot meet these specifications, because R_F must equal 1000 k. Inserting a T network with $R_2 = R_4 = 10$ k and $R_3 = 485$ k approximately meets the specifications.

3.7 Video Amplifiers

Video signals contain high frequencies, and they use coaxial cable to transmit and receive signals. The cable connecting these circuits has a characteristic impedance of 75 Ω . To prevent reflections, which cause distortion and ghosting, the input and output circuit impedances must match the 75 Ω cable.

Matching the input impedance is simple for a noninverting amplifier, because its input impedance is very high; just make $R_{\text{IN}} = 75$ Ω . R_F and R_G can be selected as high values, in the hundreds of ohms range, so that they have minimal effect on the impedance of the input or output circuit. A matching resistor, R_M , is placed in series with the op amp output to raise its output impedance to 75 Ω ; a terminating resistor, R_T , is placed at the input of the next stage to match the cable (Figure 3.9).

The matching and terminating resistors are equal in value, and they form a voltage divider of 1/2 because R_T is not loaded. Very often, R_F is selected equal to R_G so that the op amp gain equals 2. Then the system gain, which is the op amp gain multiplied by the divider gain, is equal to 1 ($2 \times 1/2 = 1$).

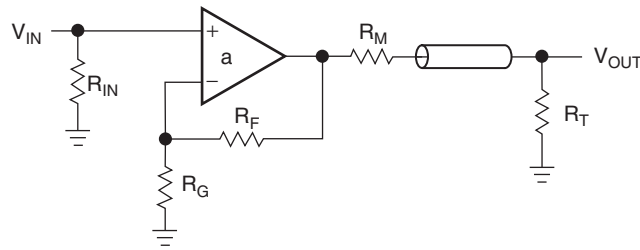


Figure 3.9: Video amplifier.

3.8 Capacitors

Capacitors are a key component in a circuit designer's tool kit; therefore, a short discussion on evaluating their affect on circuit performance is in order. Capacitors have an impedance of $X_C = 1/2\pi fC$. Note that, when the frequency is zero, the capacitive impedance (also known as *reactance*) is infinite; and when the frequency is infinite, the capacitive impedance is zero. These end points are derived from the final value theorem, and they are used to get a rough idea of the effect of a capacitor. When a capacitor is used with a resistor, they form what is called a *break point*. Without going into complicated math, just accept that the break frequency occurs at $f = 1/(2\pi R_C)$ and the gain is -3 dB at the break frequency.

The low pass filter circuit shown in Figure 3.10 has a capacitor in parallel with the feedback resistor. The gain for the low pass filter is given in Equation (3.20):

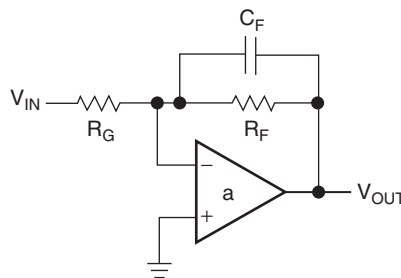


Figure 3.10: Low pass filter.

$$\frac{V_{OUT}}{V_{IN}} = -\frac{X_C || R_F}{R_G} \quad (3.20)$$

At very low frequencies $X_C \Rightarrow \infty$, so R_F dominates the parallel combination in Equation (3.20), and the capacitor has no effect. The gain at low frequencies is $-R_F/R_G$. At very high frequencies $X_C \Rightarrow 0$, so the feedback resistor is shorted out, thus reducing the circuit gain to zero. At the frequency where $X_C = R_F$, the gain is reduced by $\sqrt{2}$ because complex impedances in parallel equal half the vector sum of both impedances.

Connecting the capacitor in parallel with R_G , where it has the opposite effect, makes a high pass filter (Figure 3.11). Equation (3.21) gives the equation for the high pass filter.

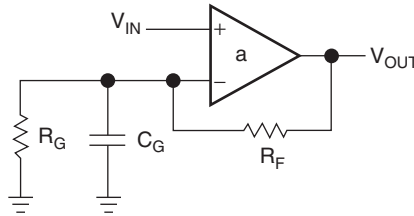


Figure 3.11: High pass filter.

$$\frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_F}{X_C || R_G} \quad (3.21)$$

At very low frequencies $X_C \Rightarrow \infty$, so R_G dominates the parallel combination in Equation (3.21), and the capacitor has no effect. The gain at low frequencies is $1 + R_F/R_G$. At very high frequencies, $X_C \Rightarrow 0$, so the gain setting resistor is shorted out, thus increasing the circuit gain to maximum.

This simple technique is used to predict the form of a circuit transfer function rapidly. Better analysis techniques are presented in later chapters for those applications requiring more precision.

3.9 Why an Ideal Op Amp Would Destroy the Known Universe

An understanding of op amp parameters, as explained in Chapter 13, is not required but would be helpful at this point. An ideal op amp has the following specifications:

- It draws no supply current and therefore has no power supplies. Hence, it doesn't even have to be turned on to be dangerous!

- It has no V_{OH} and V_{OL} limitations because it has no power supplies. Therefore, its output voltage swings from $\pm\infty$ V.
- It has zero output resistance, and therefore it is capable of supplying infinite current at each voltage extreme.
- It has infinite gain, and therefore the slightest input signal would allow it to swing to positive and negative infinite voltage (without feedback components, that is).
- It has infinite slew rate and therefore would swing to either rail—both equally destructive—instantly.

Therefore, an ideal op amp, just lying on the table with no power applied, would instantly take a quantum difference between its positive and negative terminals and amplify that difference to an infinite voltage output at infinite current. The resulting surge of power would be a sphere of destruction radiating out from the op amp at the speed of light!

This somewhat humorous analysis is included to drive home a few points:

1. If the ideal op amp model is employed, the engineer must also know how real world op amp parameters degrade and alter the ideal op amp model. An ideal op amp model is a useful tool for initial phases of simulation and analysis but does not adequately explain real world op amp behavior.
2. All of this mathematical analysis can be boiled down to a simple concept: *The op amp will do whatever it has to do at its output to equalize the voltages at its inputs.* This is the entire content of this book, distilled to its simplest form. This fundamental concept can be used to derive all of the behavior of all op amp circuits in all applications. Of course, it must be filtered through statement 1 for real world op amps.
3. A very astute analog design engineer might see one fallacy in this “death star” scenario: “Where is the return path?” When a single ended op amp is employed, the return path is to ground. But when no power supplies are utilized, there is no ground! So, to be technically correct, the only type of ideal op amp that would destroy the known universe is a fully differential type, where the return for one output is the other output. This book explains in detail the subject of proper return to ground for single ended op amps and single ended op amps configured in single supply operation.

3.10 Summary

When the proper assumptions are made, the analysis of op amp circuits is straightforward. These assumptions, which include zero input current, zero input offset voltage, and infinite gain, are realistic assumptions because the new op amps make them essentially true in most real world applications.

When the signal comprises low frequencies, the gain assumption is valid because op amps have very high gain at low frequencies. When CMOS op amps are used, the input current is in the femtoamp range; close enough to zero for most applications. Laser trimmed input circuits reduce the input offset voltage to a few microvolts; close enough to zero for most applications. The ideal op amp is becoming real, especially for undemanding applications.